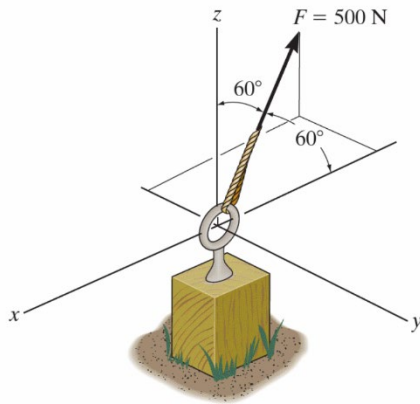


Tutorial - Vectors

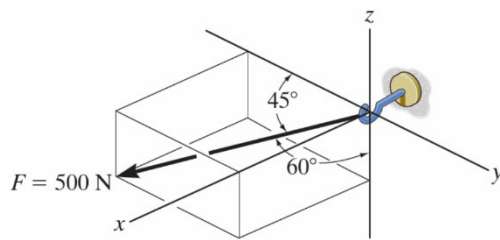
Question 1

F2-14. Express the force as a Cartesian vector.



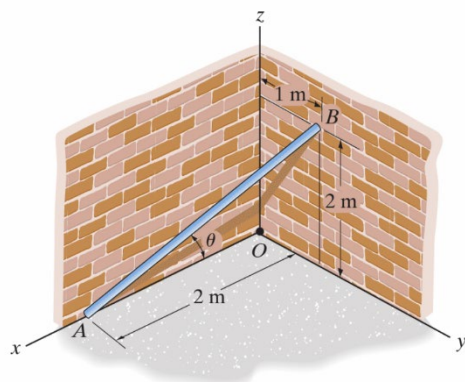
Question 2

F2-15. Express the force as a Cartesian vector.



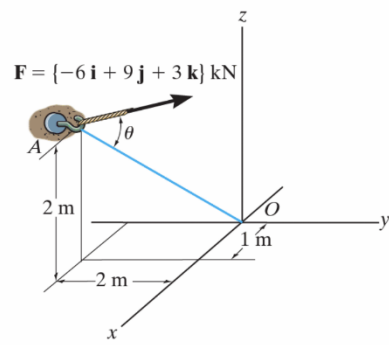
Question 3

F2-20. Determine the length of the rod and the position vector directed from A to B . What is the angle θ ?



Question 4

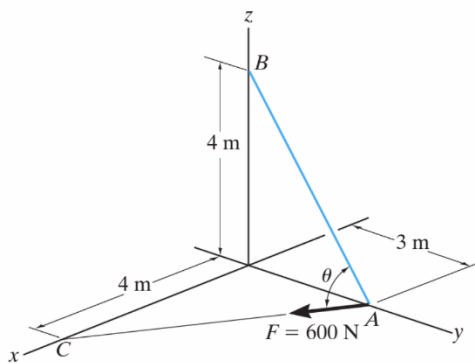
F2-25. Determine the angle θ between the force and the line AO .



Prob. F2-25

Question 5

F2-26. Determine the angle θ between the force and the line AB .



Question 1, Problem F2-14

Using the same nomenclature as in the textbook

$$\cos\alpha = \cos\gamma = \cos 60^\circ = \frac{1}{2}.$$

From the properties of direction cosines

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1.$$

Combining these two results gives

$$\cos\beta = \frac{1}{\sqrt{2}}.$$

By analogy with equation 2-9 in the statics textbook

$$\mathbf{F} = 500 \left(-\frac{1}{2}\mathbf{i} - \frac{1}{\sqrt{2}}\mathbf{j} + \frac{1}{2}\mathbf{k} \right) = 500 \left(-\frac{1}{2}\mathbf{i} + -\frac{\sqrt{2}}{2}\mathbf{j} + \frac{1}{2}\mathbf{k} \right) = 250(-\mathbf{i} - \sqrt{2}\mathbf{j} + \mathbf{k}) \text{ N}.$$

Don't forget the units! Look at the axes on the diagram to determine positive/negative vector components (an alternative method is shown in the solution to F2-15).

Question 2, Problem F2-15

Be careful – direction cosines must be measured with respect to the **positive** x, y and z axis directions!

$$\cos\beta = \cos(180^\circ - 45^\circ) = \cos(135^\circ) = \cos(90^\circ + 45^\circ) = \cos 90^\circ \cos 45^\circ - \sin 90^\circ \sin 45^\circ.$$

So

$$\cos\beta = -\sin 45^\circ = -\frac{1}{\sqrt{2}}.$$

$$\cos\gamma = \cos(180^\circ - 60^\circ) = \cos(120^\circ) = \cos(90^\circ + 30^\circ) = \cos 90^\circ \cos 30^\circ - \sin 90^\circ \sin 30^\circ.$$

So

$$\cos\gamma = -\sin 30^\circ = -\frac{1}{2}.$$

From the properties of direction cosines

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1.$$

Substituting the values for $\cos\beta$ and $\cos\gamma$, from above, into this relationship, gives

$$\cos\alpha = \frac{1}{2}.$$

By analogy with equation 2-9 in the statics textbook

$$\mathbf{F} = 500 \left(\frac{1}{2} \mathbf{i} - \frac{1}{\sqrt{2}} \mathbf{j} - \frac{1}{2} \mathbf{k} \right) = 500 \left(\frac{1}{2} \mathbf{i} - \frac{\sqrt{2}}{2} \mathbf{j} - \frac{1}{2} \mathbf{k} \right) = 250(\mathbf{i} - \sqrt{2} \mathbf{j} - \mathbf{k}) \text{ N.}$$

Don't forget the units!

Question 3 Problem F2-20

The vector position of B with respect to A is given by

$$\mathbf{r}_{AB} = -4\mathbf{i} + 2\mathbf{j} + 4\mathbf{k} \text{ m.}$$

$$\text{Its modulus is } r_{AB} = \sqrt{16 + 4 + 16} = \sqrt{36} = 6 \text{ m.}$$

We have to find the angle that $\mathbf{r}_{AB}$ makes with the negative x-axis direction, i.e. with $-\mathbf{i}$. This angle is given by

$$\cos\theta = \frac{\mathbf{r}_{AB} \cdot (-\mathbf{i})}{6 \times 1} = \frac{2}{3}.$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{2}{3}\right) = 48^\circ.$$

Question 4, Problem F2-25

The position vector of A is given by

$$\mathbf{r}_{OA} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k} \text{ m.}$$

The negative sign arises because x, y and z are considered to form a right-handed system so A is displaced in the negative y-axis direction.

$$\text{So } \mathbf{r}_{AO} = -\mathbf{i} + 2\mathbf{j} - 2\mathbf{k} \text{ m.}$$

(Note that we want the vector from A to O to find the angle required.)

$$\text{Its modulus is given by } r_{AO} = \sqrt{1 + 4 + 4} = 3 \text{ m.}$$

$$\text{The modulus of } \mathbf{F} \text{ is given by } F = \sqrt{36 + 81 + 9} = \sqrt{126} = 11.22 \text{ N.}$$

(Notice that we want to keep this result with several figures after the decimal point, to prevent rounding errors in subsequent calculations.)

$$\cos\theta = \frac{\mathbf{r}_{AO} \cdot \mathbf{F}}{33.66} = \frac{6+18-6}{33.7} = \frac{18}{33.7}.$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{18}{33.66}\right) = 57^\circ.$$

Question 5, Problem F2-26

The vector from A to B is given by

$$\mathbf{r}_{AB} = -3\mathbf{j} + 4\mathbf{k} \text{ m.}$$

Note that triangle OAC is right-angled with shortest sides of 3 and 4 m; so its longest side is 5 m.

Then the direction cosines of $\mathbf{F}$ are $4/5$, $-3/5$ (the minus sign arises because it points in the negative y-axis direction) and 0, with respect to the x, y and z-axes, respectively.

$$\text{So } \mathbf{F} = \frac{4}{5} \times 600\mathbf{i} - \frac{3}{5} \times 600\mathbf{j} = 480\mathbf{i} - 360\mathbf{j}.$$

The angle between them is given by

$$\cos\theta = \frac{(-3\mathbf{j}+4\mathbf{k}) \cdot (480\mathbf{i}-360\mathbf{j})}{5 \times 600} = \frac{3 \times 360}{5 \times 600} = \frac{36}{100}.$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{36}{100}\right) = 69^\circ.$$

If you have used column or row vectors to solve questions above, refer to section 1.8 in the lecture notes for the correct matrix notation.